Sales Performance Gap Analysis and Discount Strategy Optimization

Problem Statement

Supramart is a shopping center in Nigeria that sells various product to the public. In 2023, the sales team worked with a new suppliers to delivered 6 foreign products such as Paseo, Carretera, VTT, Amarilla, Velo and Montana within a target plan. Unfortunately, the sales team experienced significant underperformance against targets plan which brought about **what is the cause of it and how discount affects patronage.** The sales team want to know the reason behind **not reaching target** by examining **how product performed regionally** and the **influence discount had on each product**.

This analysis focuses on key metrics such as actual sales, forecast sales, Discount, product and Geographic region.

This project was carried out to understand how this different product performed and what region contributed the most.

Project Objectives

- Investigate root cause behind sales shortfall.
- How did discount affect product sales from January 2023 to April 2024?
- Identify which region contributed the most from January 2023 to April 2024.

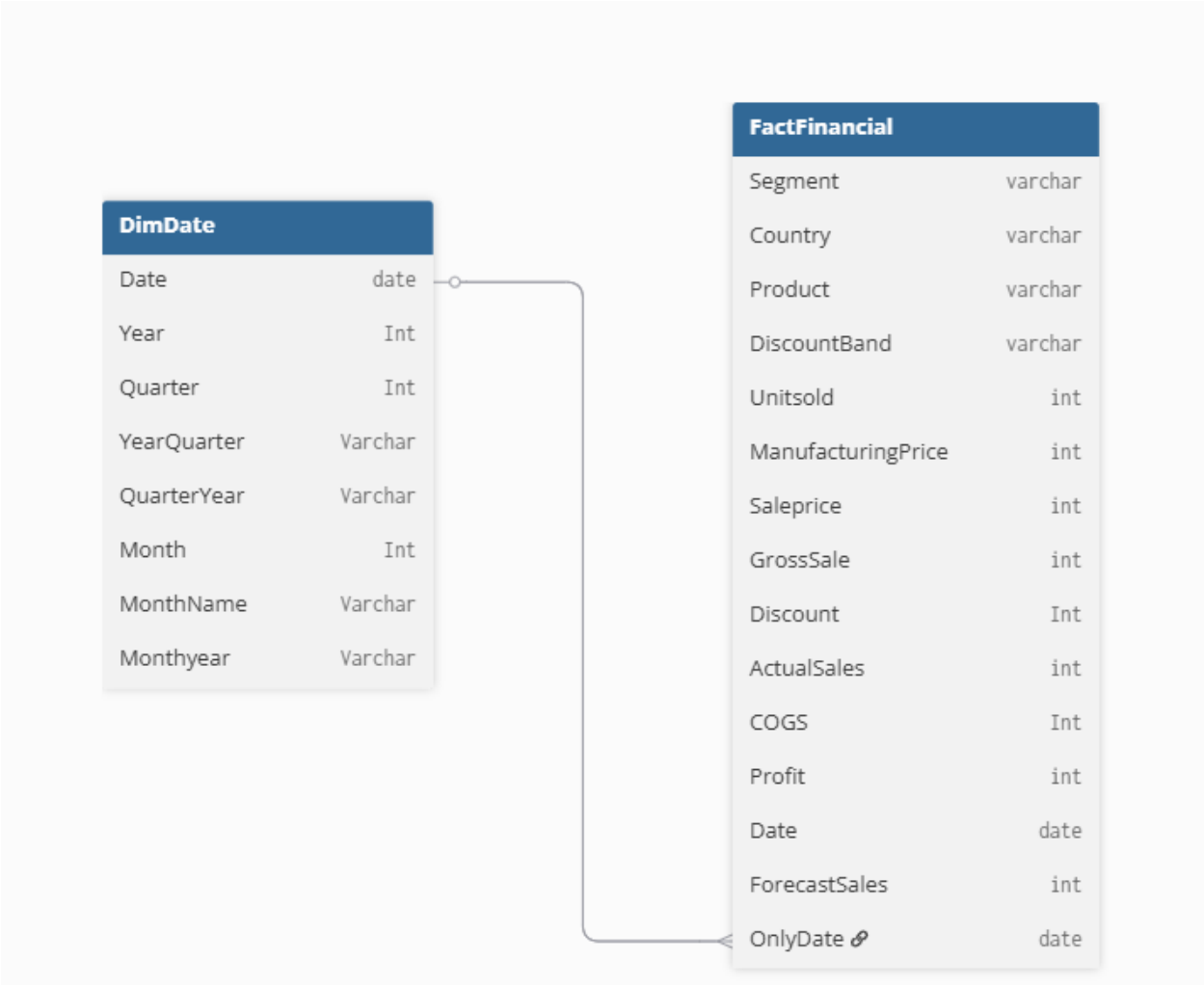
Data Structure

This Dataset contains important metrics such as actual sales, forecast sales, profit and discount from January 2023 to April 2024.

- **Factfinancial**: This dataset contains key metrics such as actual sales, forecast sales, geographical region, product, quantity sold, COGS, discount band, discount, segment etc.

Metrics calculated for this table are : profit Margin(Total profit/Total actual sales), %actual sales vs forecast sale(Total actual sales/Total forecast sales), Average profit by product AVERAGEX(ALLSELECTED[FACT_Financial' [Product], [Total Profit]],

- **DimDate**: The table contains column with day, weeks, months, month year and year.



Technical Details

This project used Power BI for all stages of data cleaning and analysis, with a focus on the root cause behind sales shortfall between January 2023 and April 2024.

Skills and Function used

Skills: Format, Calculate, Aggregate Function(Sum, Divide, AVERAGEX)

Steps Details

Initial Inspection: Checked all source tables(DimDate, Factfinancial) for missing values, duplicate entries and inconsistent data types.

Cleaning & Standardization: Renamed row.

Calculated Metrics: %Actual Sales vs Forecast Sales, profit Margin, Average profit by product.

Data Cleaning Action

Here are the key cleaning operation

- **Renamed Row**

Purpose: Rename the date column 'only date' after extracting the date from the date/time row.

Date Table Details

The date was inserted into the StartDate and the last day was inserted as the EndDate and this helped to create a date table which contains day, weeks, month, month year, Quarter year, year.

Steps Details

The date table was created through this method;

```
let

//Create parameter

    StartDate = List.Min(FACT_Financial[Only Date]),

    EndDate = List.Max(FACT_Financial[Only Date]),

    Step = Duration.Days(EndDate - StartDate)+1,

// Create the List of date

    Source = List.Dates(StartDate,Step, #duration(1,0,0,0)),

    #"Converted to Table" = Table.FromList(Source, Splitter.SplitByNothing(), null, null,
ExtraValues.Error),

    #"Changed Type" = Table.TransformColumnTypes(#"Converted to Table",{{"Column1", type
date}}),

    #"Renamed Columns" = Table.RenameColumns(#"Changed Type",{{"Column1", "Date"}}),

// Create additional columns

    #"Inserted Year" = Table.AddColumn(#"Renamed Columns", "Year", each Date.Year([Date]),
Int64.Type),

    #"Inserted Quarter" = Table.AddColumn(#"Inserted Year", "Quarter", each
Date.QuarterOfYear([Date]), Int64.Type),
```

```
    #"Inserted Quarter Year" = Table.AddColumn(#"Inserted Quarter", "Quarter Year", each Text.Combine({"Q", Text.From([Quarter], "en-US"), " ", Text.From([Year], "en-US")}), type text),

    #"Inserted Month" = Table.AddColumn(#"Inserted Quarter Year", "Month", each Date.Month([Date]), Int64.Type),

    #"Inserted Month Name" = Table.AddColumn(#"Inserted Month", "Month Name", each Text.Start(Date.MonthName([Date]),3), type text),

    #"Inserted Month Year" = Table.AddColumn(#"Inserted Month Name", "Month Year", each Text.Combine({[Month Name], " ", Text.From([Year], "en-US")}), type text),

    #"Inserted Month Year S" = Table.AddColumn(#"Inserted Month Year", "Month Year S", each Text.Combine({Text.From([Year], "en-US"), Text.PadStart(Text.From([Month], "en-US"), 2, "0")}), type text),

    #"Inserted End of Month" = Table.AddColumn(#"Inserted Month Year S", "End of Month", each Date.EndOfMonth([Date])),

    #"Inserted Week Day Name" = Table.AddColumn(#"Inserted End of Month", "Week Day", each Text.Start(Date.DayOfWeekName([Date]),3), type text),

    #"Inserted Day of Week" = Table.AddColumn(#"Inserted Week Day Name", "Day of Week", each Date.DayOfWeek([Date]), Int64.Type),

    #"Inserted Day" = Table.AddColumn(#"Inserted Day of Week", "Day", each Date.Day([Date]), Int64.Type),

    #"Inserted Week of Month" = Table.AddColumn(#"Inserted Day", "Week of Month", each Date.WeekOfMonth([Date]), Int64.Type),

    #"Inserted Week of Month W" = Table.AddColumn(#"Inserted Week of Month", "Week", each Text.Combine({"W", Text.From([Week of Month], "en-US")}), type text),

    #"Inserted Week of Year" = Table.AddColumn(#"Inserted Week of Month W", "Week of Year", each Date.WeekOfYear([Date]), Int64.Type),

    #"Inserted Week of Year W" = Table.AddColumn(#"Inserted Week of Year", "Week of Year W", each Text.Combine({"W", Text.From([Week of Year], "en-US")}), type text),

    #"Added Fiscal Month" = Table.AddColumn(#"Inserted Week of Year W", "Fiscal Month", each if Date.Month([Date]) <= 3 then Date.Month([Date]) + 9 else Date.Month([Date]) - 3),

    #"Added Fiscal Year" = Table.AddColumn(#"Added Fiscal Month", "Fiscal Year", each if Date.Month([Date]) <= 3 then Date.Year([Date]) else Date.Year([Date]) + 1),

    #"Added Fiscal Quarter" = Table.AddColumn(#"Added Fiscal Year", "Fiscal Quarter", each Number.RoundUp([Fiscal Month]/3))
```

in

#"Added Fiscal Quarter"

Executive Summary

This analysis reveals that the team overestimated the market demand with **Total Actual Sales** at **$118.7M** falling **3.4%** below **forecast at $122.9M**.Though, there is a decrease in **Total actual sales** from the previous year but **Total Profit** remains strong at **$16.9M**. **Paseo** and **VTT** products were the primary drivers of sales and profit, while **Carretera** and **Montana** are underperforming. Also, the relationship between high-discount bands and sales was not favourable which reveals that discount strategy is not functioning as an effective sales driver.

The Sales team needs to conduct market research to identify untapped opportunities and strengthen market leadership in high-performing segments .

Key Insight

- The root cause behind the $4.2m sales short fall was a clear case of **Over estimation**. The Sales team overestimated the market demand with **Total Actual Sales** at **$118.7M** falling **3.4%** below **forecast at $122.9M**.

- Paseo and VTT emerges as the top performing products in sales with $33.01M and $20.51 respectively. They had the highest profit margin (14.53%) which exceeds the average profit. The products **Carretera** and **Montana** were the main contributors to the sales shortfall. Their sales figures are lower than their forecasted targets.

- Discount strategy did not really function as an effective sales driver hence higher discount reduced COGS across all product.

- Based on the quantity sold between the Actual Sales and Forecast Sales by region, the top contributing region is the **United States of America**. This region has the highest actual sales volume followed by **Mexico** with actual sales closely exceeding the forecast then **Canada, France, and Germany** also contribute to sales but at lower volumes than the United States.

- Monthly sales data shows a strong seasonal trend, with sales peaking in **April** and **October** in both 2023 and 2024.

Recommendation

- Investigate the root cause behind sales shortfall.

Implement statistical forecasting models that incorporates seasonal factors and develop planning for different market condition.

- How did discount affect product sales from January 2023 – April 2024?

Expand medium discount initiatives across all product and implement pricing for under performing product.

- Identify which region contributed the most from January 2023 – April 2024?

Focus on maximizing opportunities in the **United States of America**, which is the top-contributing market. This could involve expanding product availability, increasing marketing spend, or launching new products tailored to this market. Investigate the reasons for the high performance in **Mexico** and consider applying similar strategies to other regions.